Anomagic: Crossmodal Prompt-driven Zero-shot Anomaly Generation

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Appendix

The supplementary material comprises the following sections, designed to offer comprehensive support and additional insights for the main manuscript:

- **Sec. A**. Details about our dataset AnomVerse
- **Sec. B**. Details about our dataset Anomagic
- **Sec. C**. More implementation and evaluation details
- **Sec. D**. More zero-shot generation results under prompts in AnomVerse
- **Sec. E**. More zero-shot generation results under user-defined prompts

A. Details about our dataset AnomVerse

The AnomVerse dataset has been meticulously curated to offer a comprehensive and diverse repository of high-quality anomaly samples tailored for zero-shot anomaly synthesis. It encompasses 12,987 images drawn from 13 publicly accessible datasets, covering an extensive array of domains such as industrial, textiles, consumer goods, medicine, and electronics. The composition of the dataset is elaborately delineated in Table 1, which specifies the total count of anomaly images, object categories, and defect types contributed by each constituent dataset. To enhance its applicability for conditioned anomaly generation, every image is paired with a descriptive caption crafted by the Doubao-Seed-1.6-thinking multi-modal large language model. Additionally, Figure 1 exemplifies anomaly-mask-caption triplets derived from AnomVerse, underscoring the dataset's versatility and practical utility.

B. Details about our method Anomagic

More results about our semantic matching process. Table 2 delineates the semantic matching procedure that correlates target objects with pre-existing anomalies within the AnomVerse dataset. For each target object, our MLLM deduces prevalent defects pertinent to its specific image category. These defects are subsequently mapped onto semantically analogous anomalies within AnomVerse, facilitating the retrieval of relevant anomaly instances across a diverse spectrum of objects and datasets. This alignment process generates anomaly-mask-caption triplets that preserve semantic coherence with the target object, thereby guaranteeing that the synthesized images maintain a high degree of fidelity to realistic anomalies with respect to semantic relevance.

More results about our contrastive anomaly mask refinement. To substantiate the proficiency of our contrastive anomaly mask refinement technique in ensuring precise alignment between synthesized anomalies and their corresponding masks, we present comparative visualizations of initial coarse masks against refined masks, accompanied by their resultant generated anomalies, as illustrated in Figure 2.

C. More implementation and evaluation details

Brief introduction to INP-Former++. To evaluate the efficacy of our anomaly generation approach in bolstering anomaly detection performance, we employ INP-Former++ (Luo et al. 2025) as our foundational model. This state-of-the-art framework integrates five critical modules: a pre-trained Encoder (Q) for robust feature extraction, an INP Extractor (E) to distill intrinsic normal prototypes from test images, a Bottleneck (B) for efficient feature compression, an INP-guided Decoder (D) that reconstructs normal tokens with reconstruction errors functioning as anomaly scores, and a Segmentation Head (H) that amplifies residual contrast to achieve precise anomaly localization. INP-Former++ exhibits exemplary performance across diverse benchmarks, including MVTec AD (Bergmann et al. 2021), VisA (Zou et al. 2022), and Real-IAD (Wang et al. 2024), excelling in unsupervised, few-shot, semi-supervised, and multi-class anomaly detection tasks. Its adaptability and robustness render it an ideal baseline for appraising the effectiveness of our synthetic anomalies across varied detection scenarios.

Details about our evaluation paradigm. For the purpose of downstream anomaly detection, we produce high-quality synthetic samples by generating one anomaly image per normal image within the testing dataset, each rendered at a resolution of 512×512 pixels. To counteract the risk of overfitting due to constrained input samples, we enforce uniform preprocessing across our method and all comparative approaches, thus cutting and pasting generated patterns to other normal images, as delineated in (Zavrtanik et al. 2021),

Table 1: **Statistical summary of AnomVerse across multiple domains.** "#Image", "#Object", and "#Defect" denote the total counts of images, object categories, and defect categories, respectively, in the corresponding dataset. Duplicate defect categories within a dataset are counted only once.

Dataset	Domain	#Image	#Object	#Defect
AITEX (Silvestre-Blanes et al. 2019)	Textiles	103	1	12
BTAD (Mishra et al. 2021)	Textiles, Industrial	290	3	1
Eyecandies (Bonfiglioli et al. 2022)	Consumer Goods	240	10	1
MPDD (Jezek et al. 2021)	Industrial	282	6	8
MVTec AD (Bergmann et al. 2021)	Industrial, Textiles, Medicine	1,258	15	48
MTD (Huang, Qiu, and Yuan 2020)	Textiles	2,676	10	6
MVTec AD-3D (Bergmann et al. 2022)	Consumer Goods, Industrial	948	12	9
VisA (Zou et al. 2022)	Consumer Goods, Industrial, Medicine	1,200	12	1
KolektorSDD2 (Božič, Tabernik, and Skočaj 2021)	Textiles	894	1	1
MulSen AD (Li et al. 2024)	Medicine, Industrial, Electronics, Consumer Goods	360	15	13
MVTec AD-2 (Heckler-Kram et al. 2025)	Textiles, Consumer Goods, Industrial	705	8	1
DAGM (für Mustererkennung 2007)	Textiles	1,054	10	1
MANTA (Fan et al. 2025)	Electronics, Medicine, Consumer Goods	2,977	38	29
Total	Industrial, Textiles, Consumer Goods, Medicine, Electronics	12,987	141	131

Table 2: **Semantic matching of target object with AnomVerse.** The MLLM analyzes each "Target Object" to identify category-specific "Common Defects," which are then matched with AnomVerse to find corresponding "Similar Defects", including their related Objects and Datasets.

Target Object	Common Defects	Similar Defects	Objects	Datasets
Candle	Deformed Wick	Deformation	Block Inductor, Led, Power Inductor	MANTA
Capsules	Scratch	Scratch	Medicine, Flat Pad, Nut	MANTA, MulSen AD
Macaroni2	Contamination	Contamination	Carrot, Bagel, Foam	MVTec AD-3D
Chewinggum	Bulge	Protrusion	Mechanics, Medicine, Led	MANTA
Pcb3	Bent Lead	Bent	Dowel, Cable Gland, Spring Pad	MVTec AD-3D, MulSen AD
Bottle	Crack	Crack	Carrot, Bagel, Cookie	MVTec AD-3D
Hazelnut	Hole	Hole	Bracket Black, Carrot, Bagel	MPDD, MVTec AD-3D
Leather	Color	Color	Foam, Capsule, Flat Pad	MVTec AD-3D, MulSen AD
Grid	Bent	Bent	Dowel, Cable Gland, Spring Pad	MVTec AD-3D, MulSen AD
Wood	Scratch	Scratch	Flat Pad, Plastic Cylinder, Toothbrush	MulSen AD

thereby enhancing sample diversity. When integrating our method with INP-Former++ for anomaly detection, only the segmentation head requires training with synthetic anomalous samples to optimize residual contrast between normal and anomalous regions, whereas the remaining components are trained solely on normal samples. Training is executed with a batch size of 16 over 20 epochs, adhering to hyperparameters consistent with the original INP-Former++ configuration to ensure a reliable and equitable performance assessment.

D. More zero-shot generation results under prompts in AnomVerse

We showcase the effectiveness of our method in zero-shot anomaly generation across a broad range of domains, including industrial (VisA, MVTec AD), medical (BraTS (de Verdier et al. 2024), OCT (Kermany et al. 2018)), and web-crawled images encompassing diverse real-world objects. To enhance computational efficiency, our experiments leverage concise captions that succinctly encapsulate object categories and defect characteristics, eschewing elaborate descriptions. The results, depicted in Figures 3 through 5, affirm the realism of the generated anomalies and highlight the method's adaptability to multi-domain contexts.

E. More zero-shot generation results under user-defined prompts

To further affirm the robustness of our method in synthesizing realistic anomalies from arbitrary user-defined prompts, we undertake extensive zero-shot evaluations under unseen prompt conditions across multiple domains. To optimize practical utility and alleviate the burden of prompt engineering, users are required to supply only concise captions. The outcomes of these experiments, illustrated in Figures 6, 7, and 8 for industrial, medical, and web-crawled real-world domains, respectively, compellingly demonstrate the superior generalization capacity of our approach without requiring additional fine-tuning. Significantly, our method adeptly accommodates prompts describing anomaly patterns absent from the training phase, thereby enabling versatile and adaptable anomaly synthesis suited to a wide array of real-world applications.

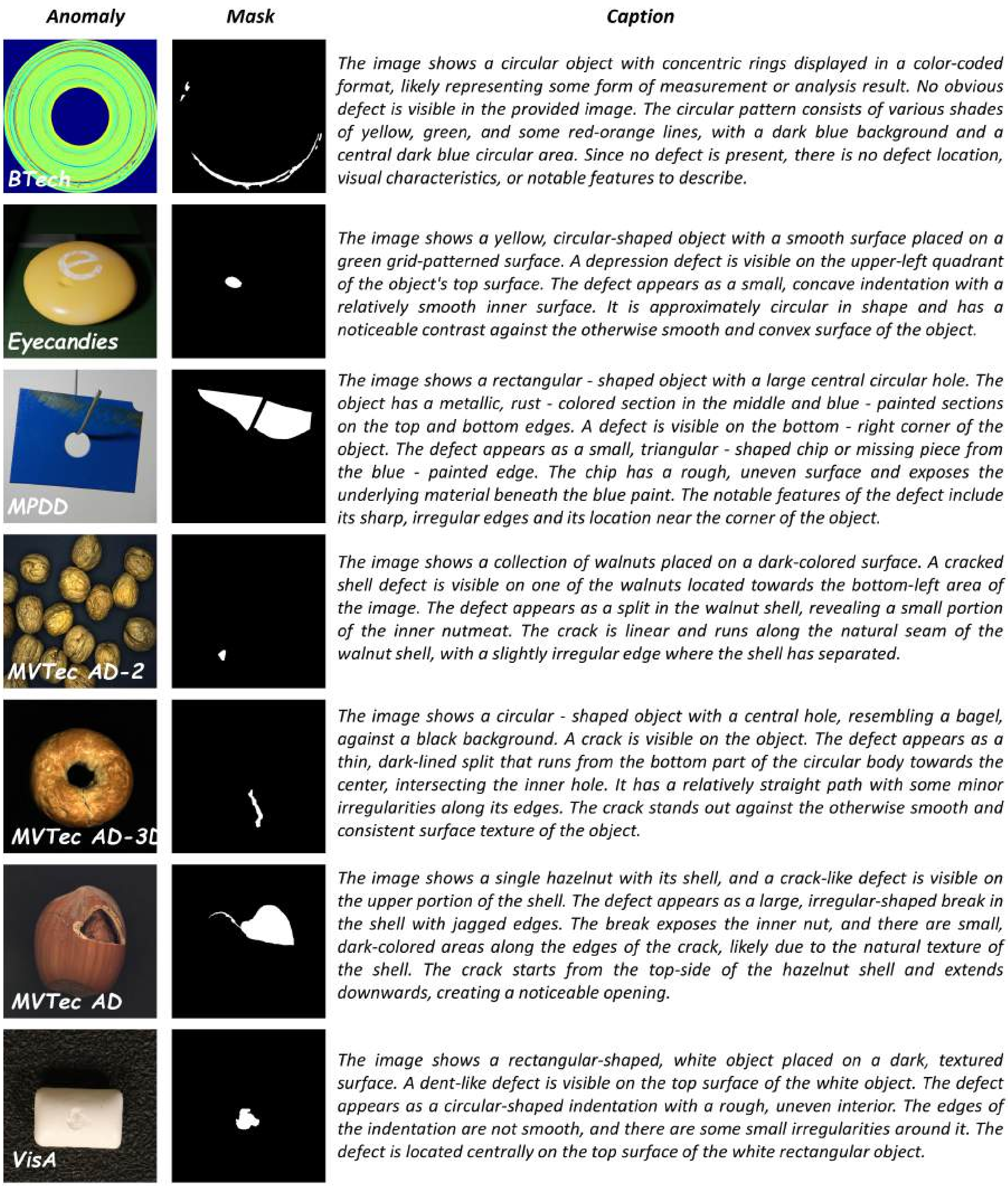

Figure 1: **Examples of anomaly-mask-caption triplets in AnomVerse.**

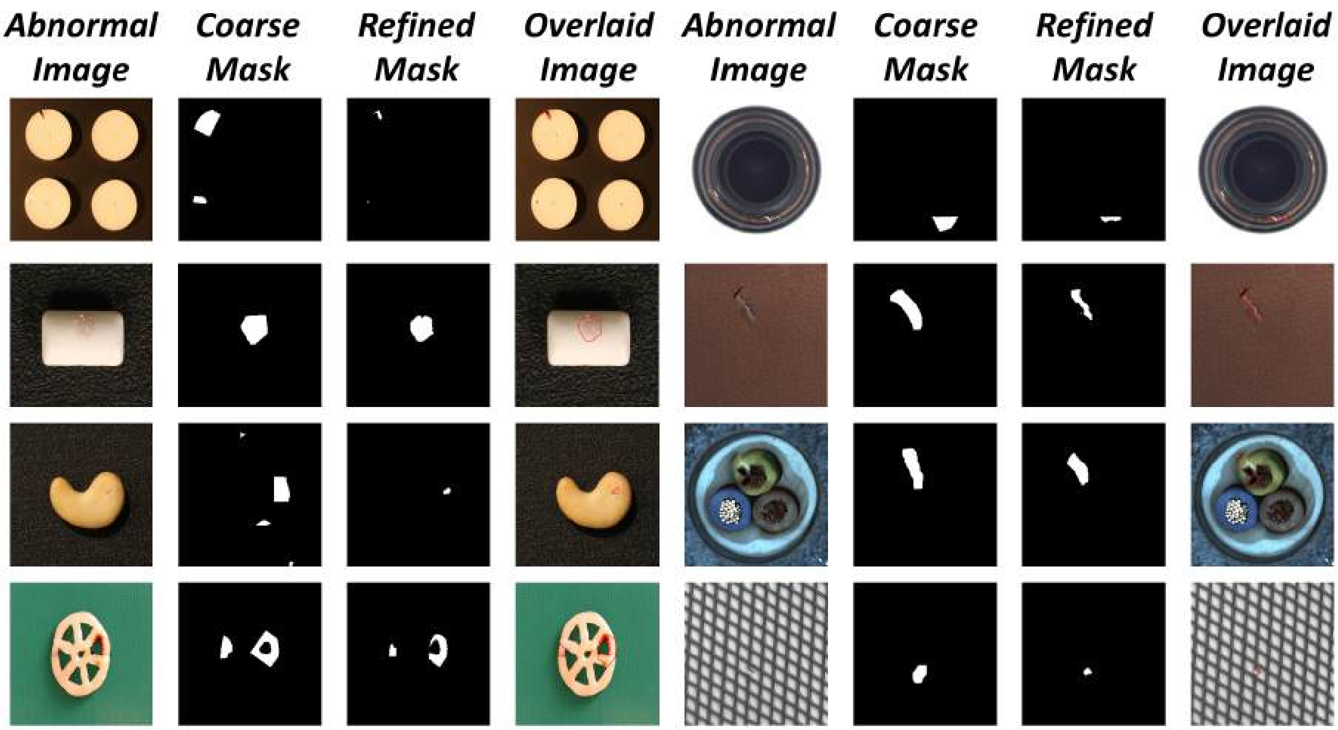

Figure 2: **Visualization of mask refinement results.** From left to right: Abnormal Image (original image with synthesized anomaly), Coarse Mask (initial mask for inpainting), Refined Mask (mask after contrastive refinement), and Overlaid Image (refined mask boundary in red overlaid on the synthesized anomaly).

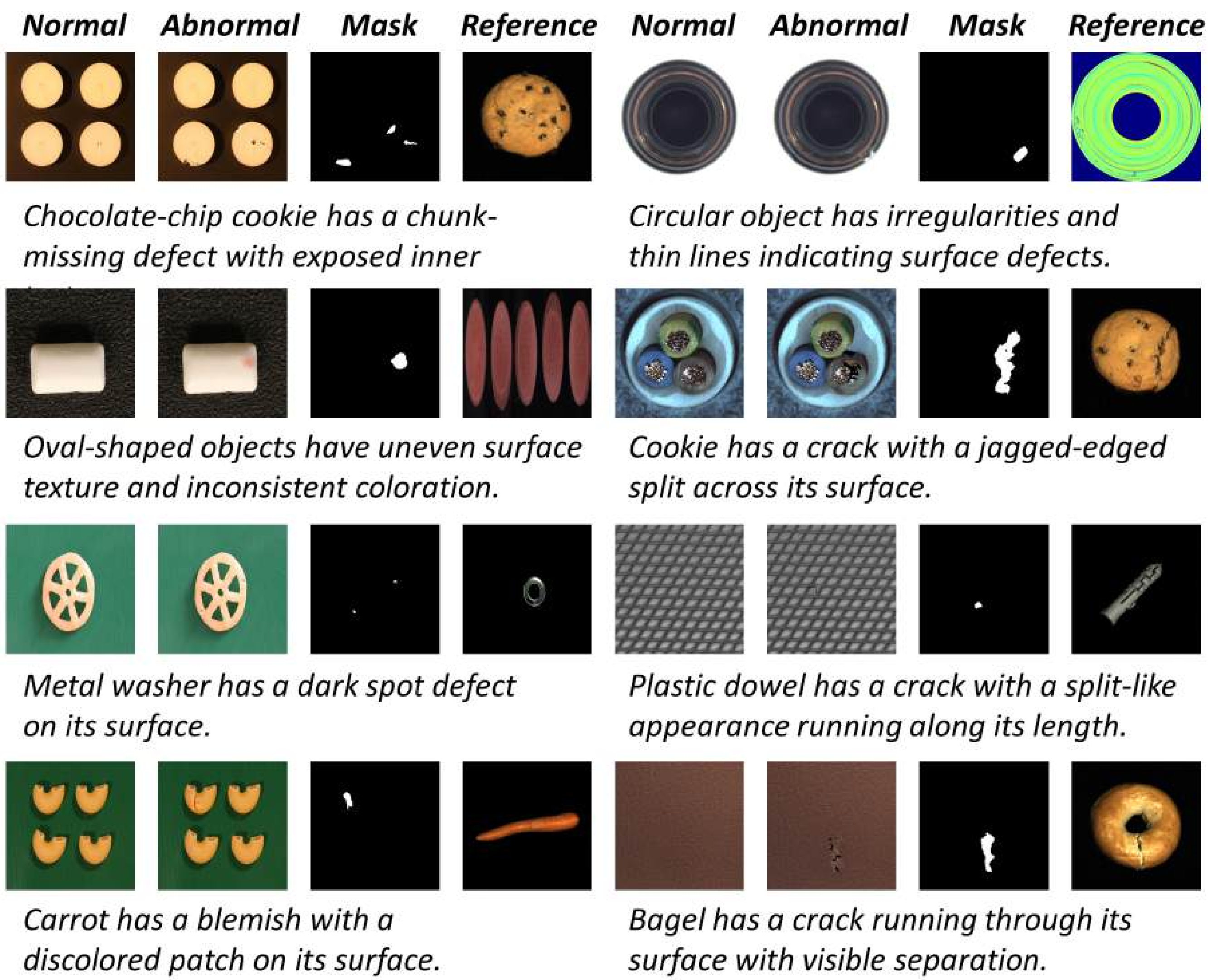

Figure 3: **Visualization of zero-shot anomaly generation results on industrial datasets MVTec AD and VisA using prompts from the AnomVerse training set.** From left to right: Normal image, Abnormal image (with synthesized anomaly), Anomaly Mask, and Reference image for anomaly synthesis.

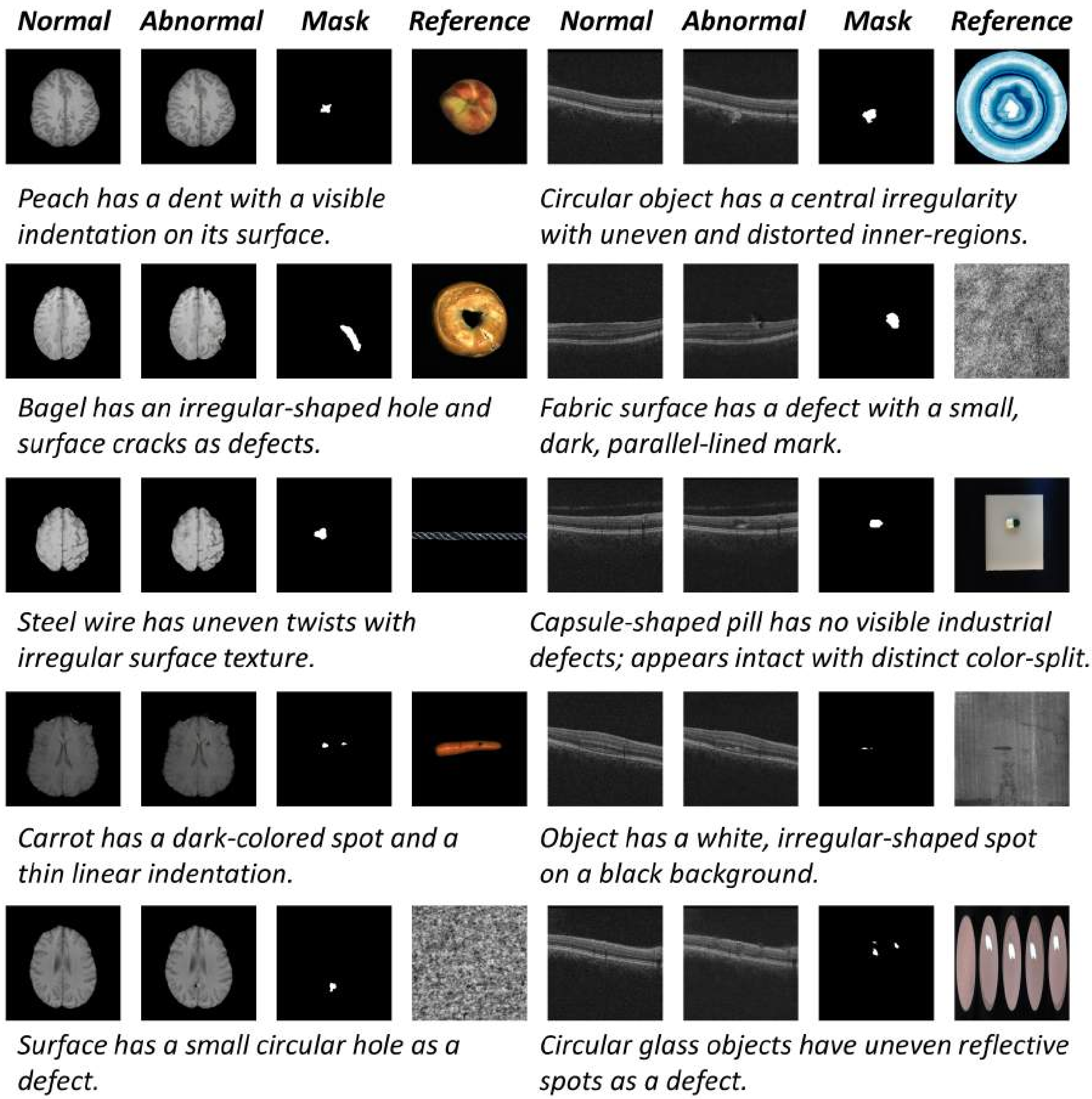

Figure 4: **Visualization of zero-shot anomaly generation results on medical datasets BraTS and OCT using prompts from the AnomVerse training set.** From left to right: Normal image, Abnormal image (with synthesized anomaly), Anomaly Mask, and Reference image for anomaly synthesis.

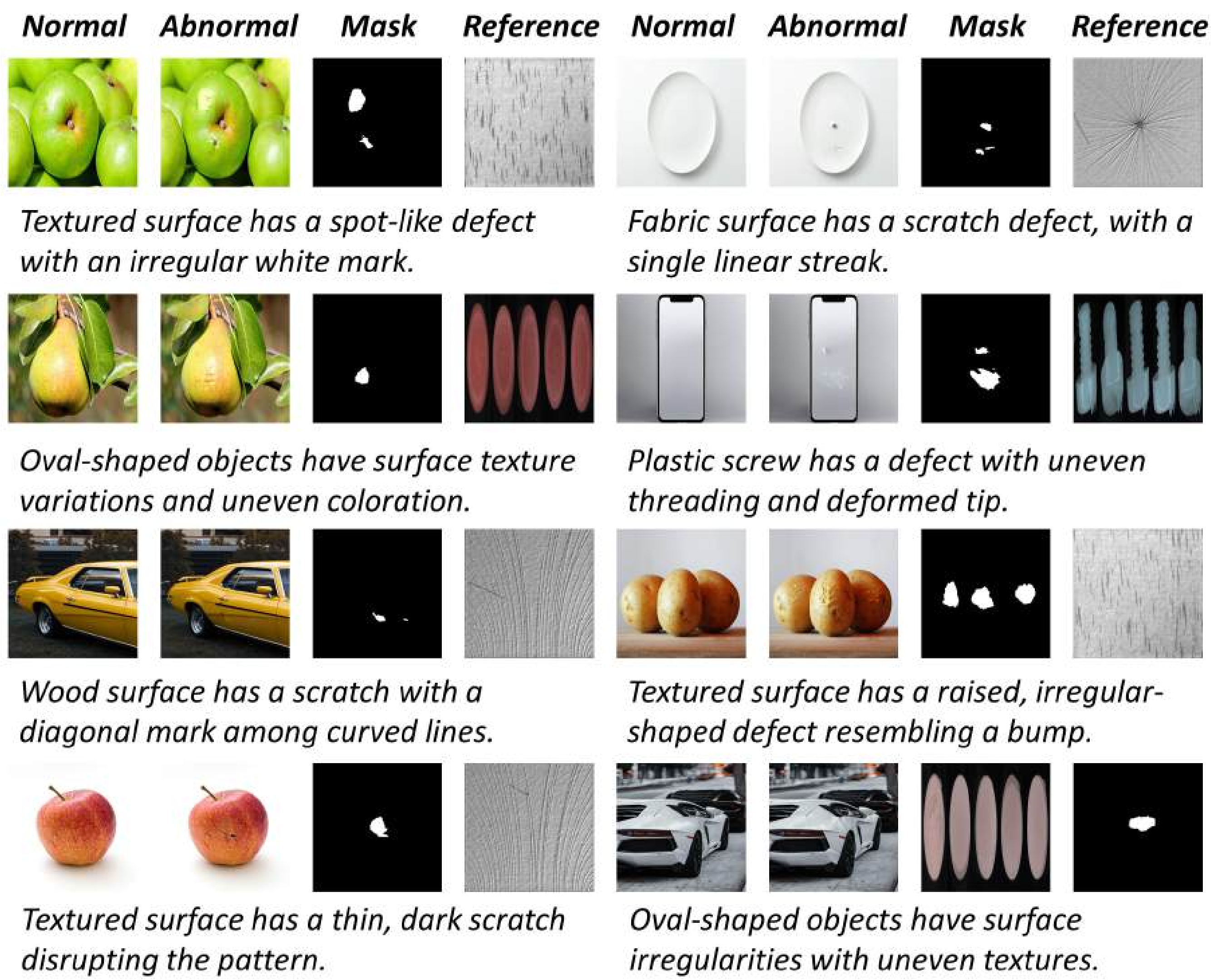

Figure 5: **Visualization of zero-shot anomaly generation results on web images using prompts from the AnomVerse training set.** From left to right: Normal image, Abnormal image (with synthesized anomaly), Anomaly Mask, and Reference image for anomaly synthesis.

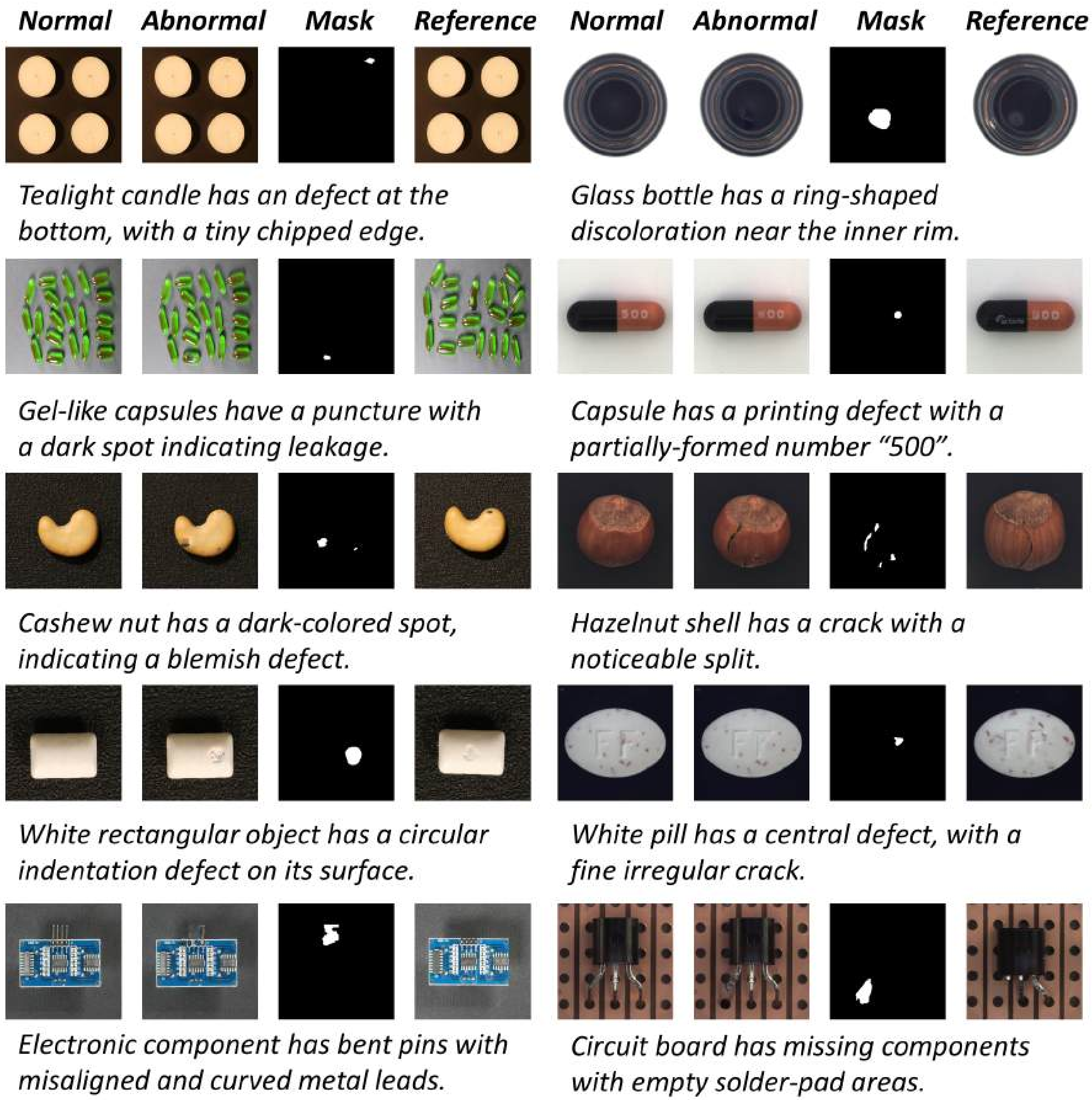

Figure 6: **Visualization of zero-shot anomaly generation results on industrial datasets MVTec AD and VisA using user-defined prompts (real anomalies from the same category).** From left to right: Normal image, Abnormal image (with synthesized anomaly), Anomaly Mask, and Reference image for anomaly synthesis.

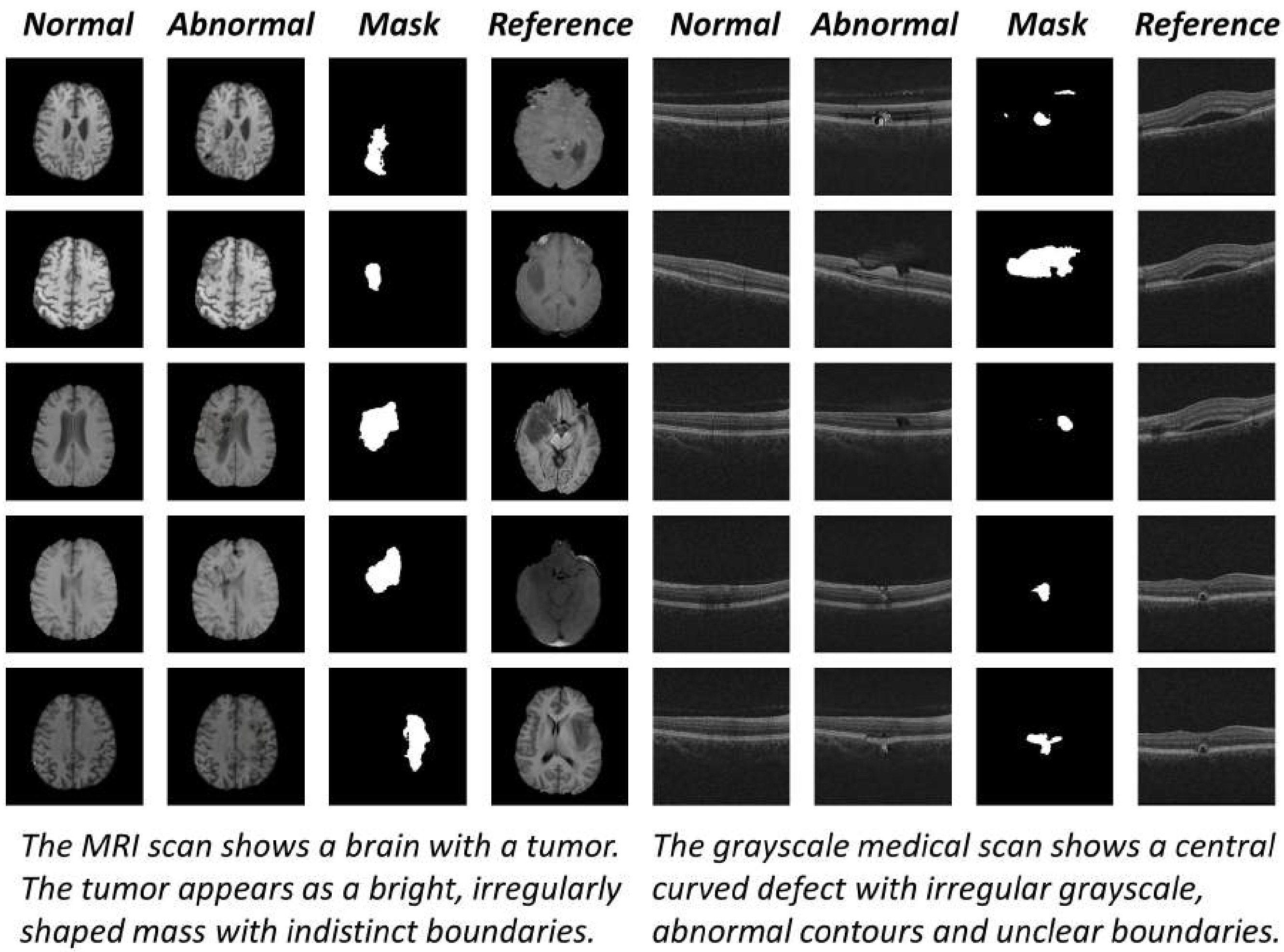

Figure 7: **Visualization of zero-shot anomaly generation results on medical datasets BraTS and OCT using user-defined prompts (real anomalies from the same category).** From left to right: Normal image, Abnormal image (with synthesized anomaly), Anomaly Mask, and Reference image for anomaly synthesis.

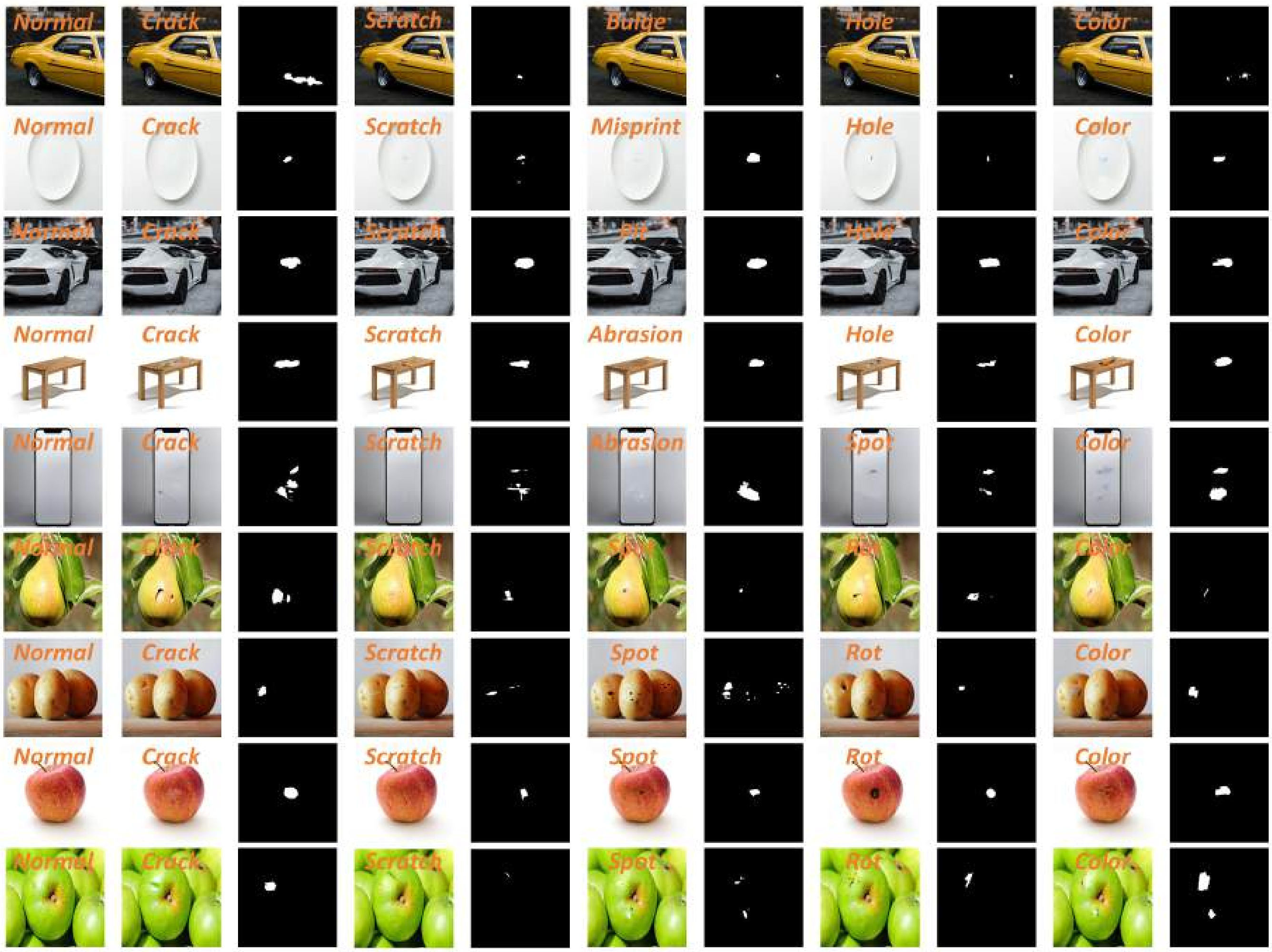

Figure 8: **Visualization of zero-shot anomaly generation results on web images using user-defined prompts.** For each object category, five distinct anomalies are synthesized with their corresponding masks.

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